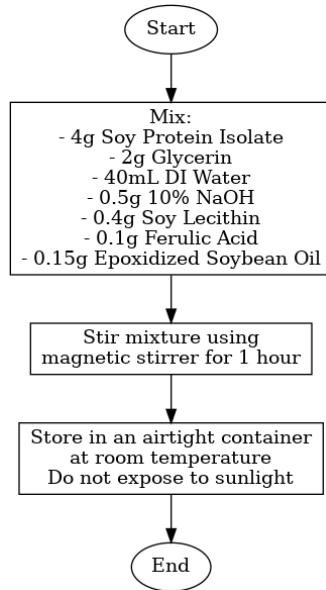


1. Critical Process Parameters and Process Flowchart

- Mixing Time & Mixing Speed: 1 hour at a consistent speed using a magnetic stirrer to disperse the ingredients evenly.
- Temperature while Mixing: Room temperature of 40 degrees Celsius to avoid denaturation of proteins..
- Drying Time & Drying Temperature: Dry at room temperature for 24 hours.



2. Critical Functionality/Performance of Soy Components

- Soy Protein Isolate (SPI):
 - Acts as a **base** for our film by binding the ingredients together in the protein matrix.
 - ProFam 781 from our Soy Kits.
- Epoxidized Soybean Oil (ESO):
 - Functions as a **cross-linking agent**, improving tensile strength of the film and alleviating any durability/longevity problems that may come up when using the SPI.
- Soy Lecithin:
 - Acts as an **emulsifier**, improving the distribution of ESO within the protein matrix.

3. Amount of Soy Components in the Product

- Soy Protein Isolate: 4g (main structural component).

- Epoxidized Soybean Oil: 0.15g (cross-linking agent).
- Soy Lecithin: 0.4g (emulsifier).
- Overall, Soy Components comprise about 93% of our product.

4. Key Technical and Commercial Issues

Technical

- Optimizing balance between SPI & ESO: Determining the correct proportion of SPI & ESO to ensure durability in the base for film.
- Optimization of Ferulic Acid Concentration: Determining the correct amount of ferulic acid to maximize the UV protection while preventing loss of film's mechanical strength.
- Film Durability: Improving tensile strength without compromising sustainability.
- Scalability: Refining the process to derive ferulic acid from soybean oil through the use of Amberlite XAD-16 for scalability.

Commercial

- Cost Competitiveness: Ensuring the product remains affordable compared to traditional UV-protective coatings.
- Incentive: Providing a strong enough incentive for people to use a more sustainable, soy-based product over more conventional alternatives.
- Education: Educating the market about the benefits of adopting a more environmentally friendly, safer alternative by demonstrating its effectiveness, sustainability, safety, and potential cost-savings.

5. Facility and Technology Requirements

- Existing Facilities
 - This product can be largely manufactured in existing facilities with the use of lab equipment and other industrial equipment (oven, stirrers, beakers, spectrophotometer, etc).
 - For scalability (and safety) purposes, we would need additional facilities to extract ferulic acid from soybean oil through the use of Amberlite XAD-16. Therefore, for this competition, we will simply buy ferulic acid. It should simply be noted that ferulic acid can be derived from soybeans as well.